

Chapter 1

Complex Numbers

1.1 Introduction

1.1.1 The Symbol j

A complex number z , is in the form of

$$z = x + jy$$

where x, y are two (real) numbers and $j = \sqrt{-1}$. $x = \text{Re}(z)$ is the real part and $y = \text{Im}(z)$ is the imaginary part of z .

Example 1.1.1

1. $z = 3 - 2j \Rightarrow \text{Re}(z) = 3, \text{Im}(z) = -2$
2. $z = 6j \Rightarrow \text{Re}(z) = 0, \text{Im}(z) = 6$
3. $z = -4 \Rightarrow \text{Re}(z) = -4, \text{Im}(z) = 0$
4. $\sqrt{25} = 5 \Rightarrow \text{Re}(\sqrt{25}) = 5, \text{Im}(\sqrt{25}) = 0$
5. $\sqrt{-25} = 5j \Rightarrow \text{Re}(\sqrt{-25}) = 0, \text{Im}(\sqrt{-25}) = 5$

1.1.2 Power of j

$$\begin{array}{llll} j & = & \sqrt{-1} & \Rightarrow j^1 = j \\ j^2 & = & -1 & \Rightarrow j^2 = -1 \\ j^3 & = & j^2 j = (-1)j & \Rightarrow j^3 = -j \\ j^4 & = & (j^2)^2 = (-1)^2 & \Rightarrow j^4 = 1 \\ j^5 & = & j^4 j = (1)j & \Rightarrow j^5 = j \end{array}$$

In general, we have

Formula 1.1.1

$$j^{4n+1} = j, \quad j^{4n+2} = -1, \quad j^{4n+3} = -j, \quad j^{4n} = 1.$$

for any positive integer n .

Next, from $j^2 = -1 \Rightarrow \frac{1}{j} = -j$

Formula 1.1.2

$$\frac{1}{j} = -j.$$

Example 1.1.2

1. $j^{42} = j^{40+2} = -1$
2. $j^{12} = 1$
3. $j^{53} = j^{52+1} = j$
4. $\frac{1}{j^{31}} = \frac{1}{j^{28+3}} = \frac{1}{-j} = j$
5. $\frac{1}{j^{50}} + j^7 = \frac{1}{-1} + (-j) = -1 - j$

1.2 Operations on Complex Numbers

1.2.1 Equality

If $z_1 = x_1 + jy_1$ and $z_2 = x_2 + jy_2$, then

$$z_1 = z_2 \iff x_1 = x_2 \text{ and } y_1 = y_2.$$

Example 1.2.1 Given $z_1 = (3a + 2) + j(2b - 1)$ and $z_2 = (b + 1) - j(a - b - 2)$. If $z_1 = z_2$

1. find a and b ;
2. write down the real and imaginary parts of z_1 and z_2 .

Solution.

1. $z_1 = z_2 \iff \left. \begin{array}{l} 3a + 2 = b + 1 \\ 2b - 1 = -(a - b - 2) \end{array} \right\} \iff \left. \begin{array}{l} 3a - b = -1 \\ a + b = 3 \end{array} \right\} \quad a = \frac{1}{2}, b = \frac{5}{2}$
2. $Re(z_1) = Re(z_2) = \quad \quad \quad Im(z_1) = Im(z_2) = \quad$

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1.2.2 Addition and Subtraction

If $z_1 = x_1 + jy_1$ and $z_2 = x_2 + jy_2$, then

$$z_1 \pm z_2 = (x_1 \pm x_2) + j(y_1 \pm y_2).$$

Example 1.2.2 Given $z = 2 + 6j$ and $w = -1 + 7j$. Find $z + w$, $2z - 3w$, $Re(-3z + w)$ and $Im(z - w)$.

Solution.

$$\begin{aligned} z + w &= (2 - 1) + (6 + 7)j \\ 2z - 3w &= (4 + 12j) - (-3 + 21j) = (4 + 3) + (12 - 21)j = \\ -3z + w &= (-6 - 18j) + (-1 + 7j) = \quad \quad \quad \Rightarrow \quad Re(-3z + w) = \\ z - w &= \quad \quad \quad \Rightarrow \quad Im(z - w) = \end{aligned}$$

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1.2.3 Multiplication

If $z_1 = x_1 + jy_1$ and $z_2 = x_2 + jy_2$, then

$$z_1 z_2 = x_1 x_2 - y_1 y_2 + j(x_1 y_2 + x_2 y_1).$$

Proof.

$$\begin{aligned} z_1 z_2 &= (x_1 + jy_1)(x_2 + jy_2) = x_1 x_2 + jx_1 y_2 + jx_2 y_1 + j^2 y_1 y_2 \\ &= x_1 x_2 - y_1 y_2 + j(x_1 y_2 + x_2 y_1) \quad (\because j^2 = -1). \end{aligned}$$

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Example 1.2.3 If $z = 2 - 3j$ and $w = 4 + 5j$, determine wz , zw , $Im(z^2)$ and $Re(w^2)$.

Solution.

$$\begin{aligned} wz &= (4 + 5j)(2 - 3j) = \\ zw &= (2 - 3j)(4 + 5j) = \\ z^2 &= (2 - 3j)^2 = \Rightarrow Im(z^2) = \\ w^2 &= (4 + 5j)^2 = \Rightarrow Re(w^2) \end{aligned}$$

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Example 1.2.4

$$(3 + 4j)(2 - 5j)(1 - 2j) =$$

1.2.4 Conjugate

If $z = x + jy$ then $\bar{z} = x - jy$ is called the conjugate of z .

Example 1.2.5

1. $z = 4 + 5j \Rightarrow \bar{z} = 4 - 5j$
2. $-2 - 3j$ and $-2 + 3j$ are conjugate pairs
3. $6 + 2j$ and $2 + 6j$ are NOT conjugate pairs
4. $5 - 3j$ and $-5 + 3j$ are NOT conjugate pairs.

Example 1.2.6 If $z = 5 + 8j$ then

1. $z + \bar{z} = (5 + 8j) + (5 - 8j) = 10 = 2Re(z)$
2. $z - \bar{z} = (5 + 8j) - (5 - 8j) = 16j = 2jIm(z)$
3. $z\bar{z} = (5 + 8j)(5 - 8j) = 25 - 40j + 40j - 64j^2 = 25 + 64.$

From the above example, we can see

Formula 1.2.1 If $z = x + jy$ then

1. $z + \bar{z} = 2x = 2Re(z)$
2. $z - \bar{z} = 2yj = 2jIm(z)$
3. $z\bar{z} = x^2 + y^2.$

1.2.5 Division

If $z_1 = x_1 + jy_1$ and $z_2 = x_2 + jy_2$, then

$$\frac{z_1}{z_2} = \frac{(x_1x_2 + y_1y_2) + j(x_2y_1 - x_1y_2)}{x_2^2 + y_2^2}.$$

Proof. Multiplying $\frac{z_1}{z_2}$ by $\frac{x_2 - jy_2}{x_2 - jy_2} \left(= \frac{\bar{z}_2}{\bar{z}_2} \right)$:

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{x_1 + jy_1}{x_2 + jy_2} = \frac{x_1 + jy_1}{x_2 + jy_2} \times \frac{x_2 - jy_2}{x_2 - jy_2} \\ &= \frac{x_1x_2 + jy_1x_2 - jy_2x_1 - j^2y_1y_2}{x_2^2 + y_2^2} = \frac{(x_1x_2 + y_1y_2) + j(x_2y_1 - x_1y_2)}{x_2^2 + y_2^2}. \end{aligned}$$

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Example 1.2.7 Given $z = 2 - 3j$, $w = 4 + 7j$, $u = 3 - 2j$ and $v = 4 - 5j$. Find $\frac{z}{w}$, $\frac{w}{z}$, $Re\left(\frac{u}{v}\right)$ and $Im\left(\frac{v}{u}\right)$.

Solution.

$$1. \quad \frac{z}{w} = \frac{2 - 3j}{4 + 7j} = \frac{2 - 3j}{4 + 7j} \frac{4 - 7j}{4 - 7j} = \frac{8 - 14j - 12j + 21j^2}{4^2 + 7^2} = \frac{-13 - 26j}{65}$$

$$2. \quad \frac{w}{z} = \frac{4 + 7j}{2 - 3j} =$$

$$3. \quad \frac{u}{v} = \frac{3 - 2j}{4 - 5j} \quad \Rightarrow \quad Re\left(\frac{u}{v}\right) =$$

$$4. \quad \frac{v}{u} = \frac{4 - 5j}{3 - 2j} \frac{3 + 2j}{3 + 2j} = \frac{22 - 7j}{13} \quad \Rightarrow \quad Im\left(\frac{v}{u}\right) =$$

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Example 1.2.8 Find the real and imaginary parts of the complex number $z + \frac{1}{z}$, for $z = \frac{2 + j}{1 - j}$.

Solution.

$$z + \frac{1}{z} = \frac{2 + j}{1 - j} + \frac{1 - j}{2 + j} =$$

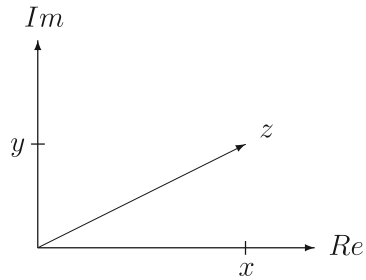
$$\therefore Re\left(z + \frac{1}{z}\right) = \quad \text{and} \quad Im\left(z + \frac{1}{z}\right) =$$

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1.3 The Argand Diagram, Modulus and Argument

1.3.1 The Argand Diagram

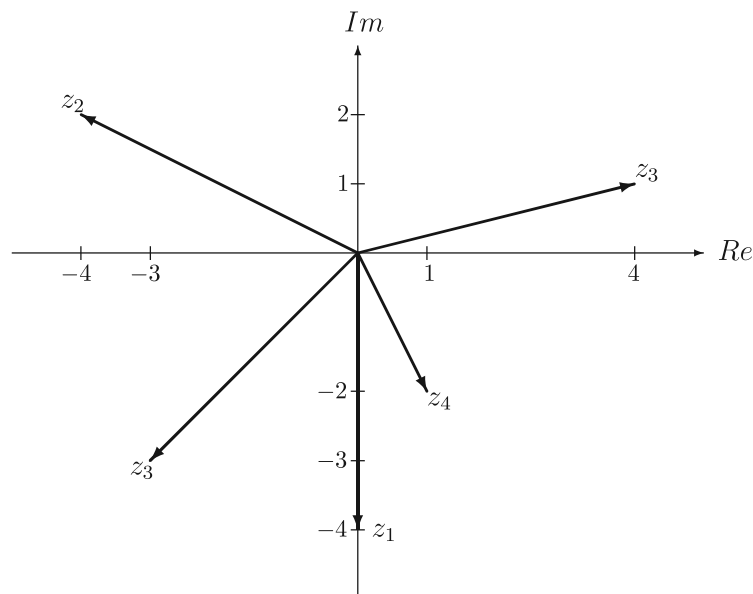
A complex number $z = x + jy$ can be represented in the xy -plane (called the Argand diagram) as follows:



The x - and y - axes are called the real and the imaginary axis respectively.

Example 1.3.1 Represent the complex numbers $z_1 = -4j$, $z_2 = -4 + 2j$, $z_3 = 4 + j$, $z_4 = 1 - 2j$ and $z_5 = -3 - 3j$ on Argand diagram.

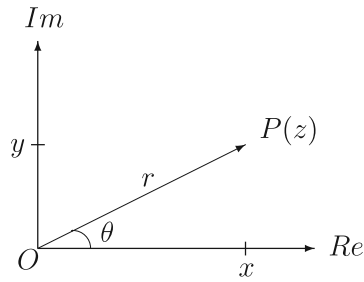
Solution.



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1.3.2 Modulus and Argument

Consider the following Argand diagram for a complex number $z = x + jy$



The length OP is called the modulus of z , (which is denoted by $|z|$ or r).

The angle between the positive real axis and OP is called the argument of z , (which is denoted by $\arg z$ or θ). From the above diagram, we have

Formula 1.3.1 For a complex number $z = x + jy$,

1. $r = |z| = \sqrt{x^2 + y^2}$
2. $\arg z = \theta$, where $\tan \theta = \frac{y}{x}$ for $x \neq 0$.

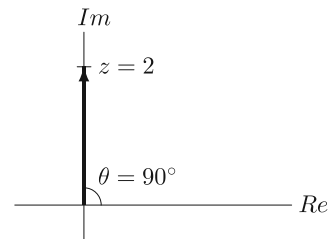
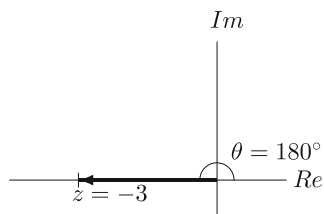
Example 1.3.2 For $z = 3 + 2j$ and $w = 5 - 7j$, we have

1. $|z| = \sqrt{3^2 + 2^2} = \sqrt{13}$
2. $|zw| = |(3 + 2j)(5 - 7j)| =$
3. $\left| \frac{z}{w} \right| = \left| \frac{3 + 2j}{5 - 7j} \cdot \frac{5 + 7j}{5 + 7j} \right| =$

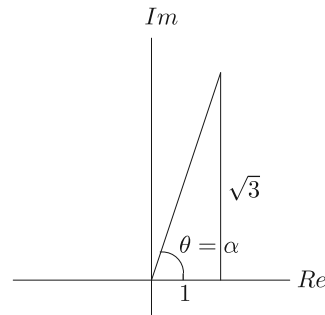
Example 1.3.3 Find the modulus and argument for -3 , $2j$, $1 + \sqrt{3}j$, $-3 - 4j$, $-5 + 2j$ and $4 - 3j$.

Solution.

1. $r = |-3| = 3$, $\theta = 180^\circ$.
2. $r = |2j| = 2$, $\theta = 90^\circ$.



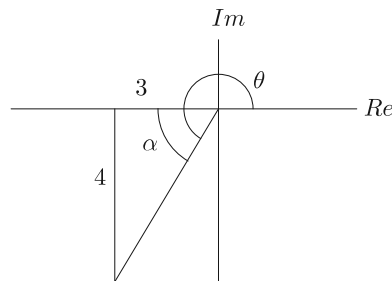
3. $r = |1 + \sqrt{3}j| = \sqrt{1 + 3} = 2$.
 $\tan \alpha = \sqrt{3} \Rightarrow \alpha = 60^\circ$.
 $\therefore$ the line in the First-Quadrant, $\theta = \alpha = 60^\circ$.



$$4. \quad r = |-3 - 4j| = \sqrt{3^2 + 4^2} = 5$$

$$\tan \alpha = \frac{4}{3} \Rightarrow \alpha = 53.13^\circ.$$

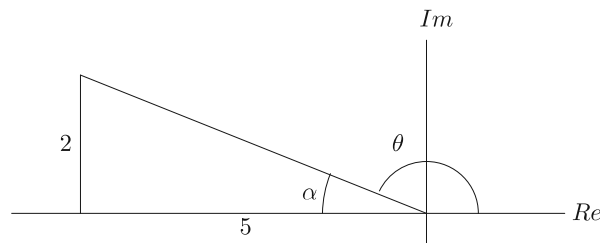
$\therefore$ the line in the Third-Quadrant, $\theta = \alpha = 180^\circ + \alpha = 233.13^\circ$ (or -126.87°)



$$5. \quad r = |-5 + 2j| = \sqrt{5^2 + 2^2} = \sqrt{29}$$

$$\tan \alpha = \frac{2}{5} \Rightarrow \alpha = 21.80^\circ.$$

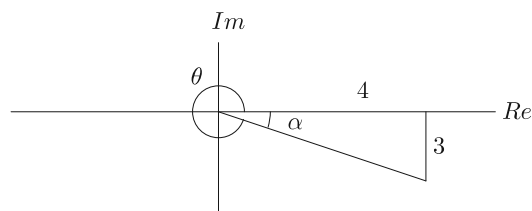
$\therefore$ the line in the Second-Quadrant, $\theta = 180^\circ - \alpha = 158.20^\circ$



$$6. \quad |4 - 3j| = \sqrt{4^2 + 3^2} = 5$$

$$\tan \alpha = \frac{3}{4} \Rightarrow \alpha = 36.87^\circ.$$

$\therefore$ the line in the Fourth-Quadrant, $\theta = \alpha = 360^\circ - \alpha = 323.87^\circ$ (or -36.13°).



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1.4 Other Forms of Complex Numbers

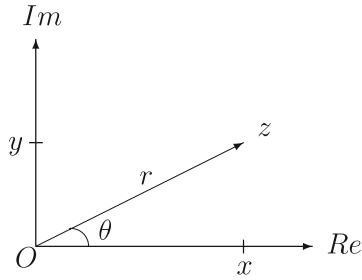
In previous section, we always write a complex number as its Cartesian form:

$$z = x + jy.$$

When solving certain problems (eg. problems involved multiplications and division), it is convenient to express a complex number in a different form (polar form or exponential form).

1.4.1 Polar Form of Complex Numbers

On the following Argand diagram for a complex number $z = x + jy$:



we have

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Since $z = x + jy$, we obtain the polar form for z :

$$z = r \cos \theta + jr \sin \theta = r(\cos \theta + j \sin \theta),$$

where r and θ can be obtained from Formula 1.3.1.

Note 1.4.1 Recall that:

$$\cos(-\theta) = \cos \theta, \quad \sin(-\theta) = -\sin \theta.$$

Example 1.4.1 Express the complex numbers in Example 1.3.3 in polar forms.

Solution.

1. $r = 3, \theta = 180^\circ \quad \therefore -3 = 3(\cos 180^\circ + j \sin 180^\circ)$
2. $r = 2, \theta = 90^\circ \quad \therefore 2j = 2(\cos 90^\circ + j \sin 90^\circ)$
3. $r = 2, \theta = 60^\circ \quad \therefore 1 + \sqrt{3}j = 2(\cos 60^\circ + j \sin 60^\circ)$
4. $r = 5, \theta = -126.87^\circ \quad \therefore -3 - 4j = 5(\cos 126.87^\circ - j \sin 126.87^\circ)$
5. $r = \sqrt{29}, \theta = 158.80^\circ \quad \therefore -5 + 2j = \sqrt{29}(\cos 158.80^\circ + j \sin 158.80^\circ)$
6. $r = 5, \theta = -36.13^\circ \quad \therefore 4 - 3j = 5(\cos 36.13^\circ - j \sin 36.13^\circ).$

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1.4.2 Exponential Form of Complex Numbers

Note 1.4.2 ¹

The Maclaurin series for e^x , $\sin x$ and $\cos x$ are given respectively by

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots \end{aligned}$$

By putting $x = j\theta$ in the Maclaurin series of e^x ,

$$\begin{aligned} e^{j\theta} &= 1 + j\theta + \frac{(j\theta)^2}{2!} + \frac{(j\theta)^3}{3!} + \frac{(j\theta)^4}{4!} + \frac{(j\theta)^5}{5!} + \dots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \dots\right) + j \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \dots\right) \\ &= \cos \theta + j \sin \theta. \end{aligned}$$

□

From Note 1.4.2, a complex number $z = r(\cos \theta + j \sin \theta)$ can also be written in the exponential form:

$$z = re^{j\theta}.$$

Note 1.4.3

In polar form $z = r(\cos \theta + j \sin \theta)$: the angle θ can be either in degrees or radians.

In exponential form $z = e^{j\theta}$: the angle θ **MUST BE** in radians.

Example 1.4.2 Express the complex numbers $-5+5j$, $-2-2j$ and $12+5j$ in its exponential form.

Solution.

1. $r = \sqrt{50}$, $\theta = 135^\circ = 0.75\pi$ $\therefore z =$.
2. $r = \sqrt{8}$, $\theta = -135^\circ = -0.75\pi$ $\therefore z =$.
3. $r = 13$, $\theta = 22.62^\circ = 0.395$ $\therefore z =$.

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Example 1.4.3

1. Express $z = 3e^{\pi j/4}$ in its polar and Cartesian forms.
2. Express $z = 5 \left(\cos \frac{\pi}{6} - j \sin \frac{\pi}{6} \right)$ in its Cartesian and exponential forms.

Solution.

¹You may skip this part.

1. Polar form: $z =$

Cartesian form: $z =$

2. Exponential form: $z =$

Cartesian form: $z =$

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1.5 Operations in Exponential and Polar Forms

1.5.1 Multiplication and Division Formulas

Note 1.5.1 Recall that:

$$e^a e^b = e^{a+b}, \quad \frac{e^a}{e^b} = e^{a-b}.$$

Let $z_1 = r_1 e^{j\theta_1}$ and $z_2 = r_2 e^{j\theta_2}$. Then

$$z_1 z_2 = (r_1 e^{j\theta_1})(r_2 e^{j\theta_2}) = r_1 r_2 e^{j(\theta_1 + \theta_2)}, \quad \frac{z_1}{z_2} = \frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}} = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)}.$$

So, in its polar forms:

$$\begin{aligned} z_1 z_2 &= r_1 r_2 [\cos(\theta_1 + \theta_2) + j \sin(\theta_1 + \theta_2)] \\ \frac{z_1}{z_2} &= \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + j \sin(\theta_1 - \theta_2)]. \end{aligned}$$

From these results, we have

Formula 1.5.1

$$\begin{aligned} z_1 z_2 &= r_1 r_2 [\cos(\theta_1 + \theta_2) + j \sin(\theta_1 + \theta_2)] = r_1 r_2 e^{j(\theta_1 + \theta_2)} \\ \frac{z_1}{z_2} &= \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + j \sin(\theta_1 - \theta_2)] = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)} \\ |z_1 z_2| &= |z_1| |z_2| = r_1 r_2, \quad \arg(z_1 z_2) = \arg(z_1) + \arg(z_2) = \theta_1 + \theta_2 \\ \left| \frac{z_1}{z_2} \right| &= \frac{|z_1|}{|z_2|} = \frac{r_1}{r_2}, \quad \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2) = \theta_1 - \theta_2. \end{aligned}$$

Example 1.5.1 For the complex numbers $z_1 = 4 \left(\cos \frac{\pi}{2} + j \sin \frac{\pi}{2} \right)$, $z_2 = 9e^{-\pi j/3}$ and $z_3 = \sqrt{2} + \sqrt{2}j$, obtain the Cartesian forms of $z_1 z_2$, z_1^2 , $\frac{z_1}{z_2}$, $\frac{1}{z_2^2}$, $z_1 z_2 z_3$ and $\frac{z_2 z_3}{z_1}$.

Solution. Given

$$z_1 = 4e^{\pi j/2}, \quad z_2 = 9e^{-\pi j/3}, \quad z_3 = 2e^{\pi j/4}.$$

1. $z_1 z_2 = (4e^{\pi j/2})(9e^{-\pi j/3}) = 36e^{\pi j/6} = 36 \left(\cos \frac{\pi}{6} + j \sin \frac{\pi}{6} \right) = 36 \left(\frac{\sqrt{3}}{2} + \frac{1}{2}j \right)$
 $= 18\sqrt{3} + 18j$
2. $z_1^2 = (4e^{\pi j/2})^2 = 16e^{\pi j} = -16$
3. $\frac{z_1}{z_2} = \frac{4e^{\pi j/2}}{9e^{-\pi j/3}} = \frac{4}{9}e^{5\pi j/6} = \frac{4}{9} \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}j \right) = -\frac{2\sqrt{3}}{9} + \frac{2}{9}j$
4. $\frac{1}{z_2^2} = \frac{1}{(9e^{-\pi j/3})^2} = \frac{1}{81}e^{2\pi j/3} = \frac{1}{81} \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}j \right) = -\frac{1}{162} + \frac{\sqrt{3}}{162}j$
5. $z_1 z_2 z_3 = (4e^{\pi j/2})(9e^{-\pi j/3})(2e^{\pi j/4}) = 72e^{5\pi j/12} = 72(0.2588 + 0.9660j)$
6. $\frac{z_2 z_3}{z_1} = \frac{(9e^{-\pi j/3})(2e^{\pi j/4})}{4e^{\pi j/2}} = \frac{9}{2}e^{-\pi j/6} = \frac{9\sqrt{3}}{4} - \frac{9}{4}j.$

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1.6 De Moivre's Theorem

1.6.1 Power of Complex Numbers

Let $z_1 = r_1(\cos \theta_1 + j \sin \theta_1)$, $z_2 = r_2(\cos \theta_2 + j \sin \theta_2)$, $\dots$, $z_n = r_n(\cos \theta_n + j \sin \theta_n)$.
 By applying Formula 1.5.1,

$$z_1 z_2 \cdots z_n = r_1 r_2 \cdots r_n [\cos(\theta_1 + \theta_2 + \cdots + \theta_n) + j \sin(\theta_1 + \theta_2 + \cdots + \theta_n)]. \quad (1.1)$$

If $z_1 = z_2 = \cdots = z_n = z$, then

$$r_1 = r_2 = \cdots = r_n = r, \quad \theta_1 = \theta_2 = \cdots = \theta_n = \theta.$$

So the equation (1.1) becomes

Formula 1.6.1 (De Moivre's Theorem)

For $z = r(\cos \theta + j \sin \theta)$ and integer $n \geq 0$,

$$z^n = r^n [\cos n\theta + j \sin n\theta] = r^n e^{jn\theta}.$$

Example 1.6.1 Calculate $(\sqrt{3} - j)^6$, $(1 - j)^{12}$ and $(-\sqrt{3} + 3j)^5$. Write your answers in Cartesian form.

Solution.

1. Since $\sqrt{3} - j = 2 \left(\cos \frac{11\pi}{6} + j \sin \frac{11\pi}{6} \right)$, [EXERCISE.]

$$(\sqrt{3} - j)^6 = \left[2 \left(\cos \frac{11\pi}{6} + j \sin \frac{11\pi}{6} \right) \right]^6 =$$

2. Since $1 - j = \sqrt{2} \left(\cos \frac{\pi}{4} - j \sin \frac{\pi}{4} \right)$, [EXERCISE.]

$$(1 - j)^{12} = \left[\sqrt{2} \left(\cos \frac{\pi}{4} - j \sin \frac{\pi}{4} \right) \right]^{12} = 64(\cos 3\pi - j \sin 3\pi) =$$

3. Since $-\sqrt{3} + 3j = \sqrt{12} \left(\cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3} \right)$, [EXERCISE.]

$$\begin{aligned} (-\sqrt{3} + 3j)^5 &= \left[\sqrt{12} \left(\cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3} \right) \right]^5 = 288\sqrt{3} \left(\cos \frac{10\pi}{3} + j \sin \frac{10\pi}{3} \right) \\ &= 288\sqrt{3} \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right) = \end{aligned}$$

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1.6.2 Roots of Complex Numbers

Let $z = r(\cos \theta + j \sin \theta)$. Write $z^{1/n} = \rho(\cos \omega + j \sin \omega)$, where n is a positive integer. Then

$$\begin{aligned} r(\cos \theta + j \sin \theta) &= z = [\rho(\cos \omega + j \sin \omega)]^n \\ &= \rho^n(\cos n\omega + j \sin n\omega). \end{aligned}$$

Comparing L.H.S and R.H.S,

$$\Rightarrow \begin{cases} \rho^n = r \\ n\omega = \theta + 2k\pi, \quad k = 0, \pm 1, \pm 2, \dots \end{cases} \quad (1.2)$$

$$\Rightarrow \begin{cases} \rho = r^{1/n} \\ \omega = \frac{\theta + 2k\pi}{n}, \quad k = 0, 1, 2, \dots, n-1. \end{cases} \quad (1.3)$$

Note 1.6.1

1. We need to “add” $2k\pi$ in the second equation in (1.2) because of

$$\cos \theta = \cos(\theta + 2k\pi) \quad \text{and} \quad \sin \theta = \sin(\theta + 2k\pi).$$

2. The reason of choosing the range of k from “0 to $n-1$ ” is explained in Note 1.6.2.

From the equations (1.3), we have

Formula 1.6.2 For $z = r(\cos \theta + i \sin \theta)$ and positive integer n ,

$$z^{1/n} = r^{1/n} \left(\cos \frac{\theta + 2k\pi}{n} + j \sin \frac{\theta + 2k\pi}{n} \right), \quad k = 0, 1, 2, \dots, n-1.$$

Example 1.6.2 Find $(-1 + \sqrt{3}j)^{1/2}$ and $(1 + j)^{1/4}$.

Solution.

$$1. z = -1 + \sqrt{3}j = 2 \left(\cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3} \right).$$

$$\begin{aligned} \Rightarrow z^{1/2} &= \sqrt{2} \left[\cos \frac{1}{2} \left(\frac{2\pi}{3} + 2k\pi \right) + j \sin \frac{1}{2} \left(\frac{2\pi}{3} + 2k\pi \right) \right], \quad k = 0, 1 \\ &= \sqrt{2} \left[\cos \left(\frac{\pi}{3} + k\pi \right) + j \sin \left(\frac{\pi}{3} + k\pi \right) \right], \quad k = 0, 1 \\ &= \sqrt{2} \left(\cos \frac{\pi}{3} + j \sin \frac{\pi}{3} \right) \quad \text{or} \quad \sqrt{2} \left(\cos \frac{4\pi}{3} + j \sin \frac{4\pi}{3} \right). \end{aligned}$$

$$2. z = 1 + j = 2^{1/2} \left(\cos \frac{\pi}{4} + j \sin \frac{\pi}{4} \right).$$

$$\begin{aligned} \Rightarrow z^{1/4} &= \\ &= \\ &= \end{aligned}$$

■

Note 1.6.2 In Part 2 of Example 1.6.2:

When $k = 4$, we get:

$$2^{1/8} \left(\cos \frac{33\pi}{16} + j \sin \frac{33\pi}{16} \right) = 2^{1/8} \left(\cos \frac{\pi}{16} + j \sin \frac{\pi}{16} \right) = \text{First Answer.}$$

When $k = 5$, we get:

$$2^{1/8} \left(\cos \frac{41\pi}{16} + j \sin \frac{41\pi}{16} \right) = 2^{1/8} \left(\cos \frac{9\pi}{16} + j \sin \frac{9\pi}{16} \right) = \text{Second Answer.}$$

$$\vdots$$

Therefore, other value of k will give the same answers.

Example 1.6.3 Solve the equation:

$$1. z^5 = 32 \quad 2. z^2 = 1 + 2j.$$

Solution.

$$1. z^5 = 32 = 32(\cos 0 + j \sin 0)$$

$$\begin{aligned} \therefore z &= [32(\cos 0 + j \sin 0)]^{1/5} = 32^{1/5} \left[\cos \frac{0 + 2k\pi}{5} + j \sin \frac{0 + 2k\pi}{5} \right] \\ &= 2 \left[\cos \frac{2k\pi}{5} + j \sin \frac{2k\pi}{5} \right], \quad k = 0, 1, 2, 3, 4. \end{aligned}$$

$$\begin{aligned}
\therefore z_1 &= 2[\cos 0 + j \sin 0] = 2 \\
z_2 &= 2 \left[\cos \frac{2\pi}{5} + j \sin \frac{2\pi}{5} \right] \\
z_3 &= 2 \left[\cos \frac{4\pi}{5} + j \sin \frac{4\pi}{5} \right] \\
z_4 &= 2 \left[\cos \frac{6\pi}{5} + j \sin \frac{6\pi}{5} \right] \\
z_5 &= 2 \left[\cos \frac{8\pi}{5} + j \sin \frac{8\pi}{5} \right]
\end{aligned}$$

$$2. \ z^2 = 1 + 2j =$$

$$\therefore z =$$

■

1.7 Loci in the Complex Plane

A *locus* (plural *loci*) is the set of points that have a specified property. In particular, a locus can be described by a graph in the Argand diagram or a Cartesian equation of x and y .

Note 1.7.1 (The equations of some common curves)

Recall that the equation of:

1. a straight line is $ax + by = c$
2. a circle is $(x - h)^2 + (y - k)^2 = R^2$
3. an ellipse is $\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1.$

1.7.1 Straight Lines

Example 1.7.1 Describe the locus of z given by

$$1. \operatorname{Re}(z) = 4 \quad 2. \arg(z - 1 - j) = \frac{\pi}{4} \quad 3. \left| \frac{z - 2j}{z - 1} \right| = 1 \quad 4. \operatorname{Im}((1 - 2j)z) = 3.$$

Solution. Write $z = x + jy$:

1. $\operatorname{Re}(z) = 4 \iff x = 4.$
 $\therefore$ the locus is a straight line.

$$2. \quad z - 1 - j = (x + yj) - 1 - j = (x - 1) + (y - 1)j$$

$$3. \quad \left| \frac{z - 2j}{z - 1} \right| = 1 \iff |z - 2j| = |z - 1| \iff |x + (y - 2)j| = |(x - 1) + yj|$$

$\therefore$ the locus is a straight line.

$$4. \quad (1 - 2j)z = \dots$$

■

1.7.2 Circles

Example 1.7.2 Find the locus of z of $|z - (2 + 3j)| = 7$.

Solution. Write $z = x + yj$.

$$\begin{aligned} |z - (2 + 3j)| = 7 &\iff |(x - 2) + (y - 3)j| = 7 \\ \sqrt{(x - 2)^2 + (y - 3)^2} &= 7 \\ (x - 2)^2 + (y - 3)^2 &= 7^2. \end{aligned}$$

$\therefore$ the locus is a circle with radius 7 and centre $(2, 3)$.

■

In general, we have

Theorem 1.7.1 The circle of radius R and centre (a, b) on the Argand diagram is represented by the equation:

$$|z - (a + bj)| = R.$$

Example 1.7.3 Find the Cartesian equation of the circle:

$$1. \quad |z + 3 - 2j| = 2 \quad 2. \quad |z + 3j| = 5.$$

Solution. Write $z = x + yj$

$$1. \quad |z + 3 - 2j| = 2 \iff |(x + 3) + (y - 2)j| = 2 \iff (x + 3)^2 + (y - 2)^2 = 2^2.$$

$\therefore$ the Cartesian equation is

$$(x + 3)^2 + (y - 2)^2 = 2^2 \quad \text{or} \quad x^2 + y^2 + 6x - 4y + 9 = 0.$$

Note 1.7.2 *It is a circle of radius 2 and centre $(-3, 2)$.*

2. The Cartesian equation of $|z + 3j| = 5$ is

$$x^2 + (y + 3)^2 = 5^2 \quad \text{or} \quad x^2 + y^2 + 6y - 16 = 0.$$

Note 1.7.3 *It is a circle of radius 5 and centre $(0, -3)$.*

■

Example 1.7.4 Find the Cartesian equation of the curve whose equation on the Argand diagram is $\left| \frac{z - j}{z - 1 - 2j} \right| = \sqrt{2}$.

Solution. Write $z = x + yj$.

■

Example 1.7.5 Find the locus of z in the Argand diagram such that $\operatorname{Re} \left(\frac{z - j}{z + 1} \right) = 0$.

Solution. Write $z = x + yj$.

$$\frac{z - j}{z + 1} = \frac{x + (y - 1)j}{(x + 1) + yj} =$$

■

1.7.3 More General Loci

Example 1.7.6 Find the Cartesian equation of the locus of z given by $|z + 1| + |z - 1| = 4$.

Solution. Write $z = x + yj$.

$$|z + 1| = 4 - |z - 1|$$